

Supplementary Appendix to “User-Centric Optimized Hybrid Electric Vehicle Charging Infrastructure: A Stackelberg Game-Based Solution”

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ABSTRACT

This Supplementary Appendix provides the materials that support the main manuscript. Section S1 states model boundaries and core assumptions. Section S2 derives the economic framework (capital-recovery factor and levelized cost of charging), defines break-even thresholds for MCV sessions and FCS utilization (Eqs. S1–S8), and instantiates them under the scenarios used in the paper (Tables S2–S3). S3 details the simulation environment, inputs, and assets, including initialization logic and logging. S4 presents the runtime symbols and complexity analysis that justify online feasibility. Unless noted, metrics follow the main text’s convention of mean over 10 independent runs.

S1. Model Formulation Boundaries and Assumptions

Table S1 defines the mathematical boundaries and key assumptions underlying our optimization formulations.

These boundaries define the theoretical framework ensuring mathematical soundness and computational tractability. The assumptions represent standard practices in operations research and game theory, providing a robust foundation for strategic analysis while acknowledging idealized conditions that facilitate optimization algorithm performance.

S2. Economic feasibility: derivations and background

This appendix provides the formulas and definitions used to produce the break-even checks in the main text. We follow standard levelized cost accounting and capital recovery practice for energy infrastructure [1, 2, 3].

S2.1. Economic Feasibility

We formalize when it is economically preferable to dispatch a MCV rather than add one more FCS port. The core objects are the capital-recovery factor (CRF), the levelized cost of charging (LCOC) for each asset, two testable thresholds, and a simple time-of-use (TOU) scaling that maps the 8 h simulation window to a representative day. Here, CRF is a standard formula that converts a one-time investment into an equivalent annual cost over the asset’s life. LCOC represents the average lifetime cost of delivering one kWh, after accounting for all capital, operating, and energy costs. TOU scaling reflects that urban demand is not constant but follows daily patterns of peaks, normal hours, and off-peaks, allowing us to extend an 8 h simulation to a realistic 24 h day.

Turning one-time costs into annual costs: A large upfront investment (CAPEX) is converted into an equivalent

annual cost using the standard capital recovery factor:

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (\text{S1})$$

with *real* discount rate r and economic life n years [3]. Here $r \in (0, 1)$ is the inflation-free rate used to convert a stream of future cash flows into an equivalent annual charge today; in practice it should reflect the operator’s real cost of capital (real Weighted Average Cost of Capital) for the charging business.

The levelized cost per kWh for one MCV is

$$\text{LCOC}_{\text{MCV}} = \frac{\text{CRF } C_{\text{MCV}} + F_{\text{MCV}} + c_{\text{km}} d_{\text{year}}}{E_{\text{year}}^{\text{MCV}}} + c_e + c_{\text{deg}} \quad (\text{S2})$$

and for one *incremental* FCS fast port is

$$\text{LCOC}_{\text{FCS}} = \frac{\text{CRF } C_{\text{FCS}} + F_{\text{FCS}}}{E_{\text{year}}^{\text{FCS}}} + c_e \quad (\text{S3})$$

following EV charging LCOC practice [1, 2].

The main cost components in these expressions are defined as: C_{MCV} and C_{FCS} are one-time capital expenditures (CAPEX): purchase and installation of the asset (e.g., for MCV the vehicle chassis, on-board DC charger, battery and thermal systems; for FCS the charger cabinet, power modules, transformer, civil works, and permitting). F_{MCV} and F_{FCS} are fixed operating expenditures (fixed OPEX) paid each year regardless of throughput (e.g., preventive maintenance, insurance, software/cloud and communications, site lease or depreciation-like site fees, back-office). Variable operating costs are modeled separately: the energy cost c_e [\$/kWh], battery degradation c_{deg} [\$/kWh] for storage-backed MCVs, and routing cost c_{km} [\$/km] multiplied by annual distance d_{year} [km]. $E_{\text{year}}^{\text{MCV}}$ and $E_{\text{year}}^{\text{FCS}}$ are the annual energies delivered [kWh]. This separation matches EV LCOC practice, where fixed costs enter through the numerator and utilization (annual kWh) drives the denominator. This means the annualized costs are divided by the actual

ORCID(s):

Table S1

Model Boundaries and Formulation Assumptions

Category	Specification	Rationale
Mathematical Model Boundaries		
Game Structure	Single-leader (operator), multiple-follower (EVs) Stackelberg game	Captures realistic market hierarchy with sequential decision-making
Decision Variables	Binary deployment $x_{im} \in \{0, 1\}$; Continuous pricing $p_m \in \mathbb{R}^+$	Mixed-integer formulation balances discrete deployment with flexible pricing
Equilibrium Concept	Stackelberg equilibrium with 2% optimality tolerance	Accounts for stochastic demand while ensuring solution stability
Optimization Horizon	Single-period with rolling updates every 5 minutes	Balances computational tractability with real-time responsiveness
Key Formulation Assumptions		
Agent Behavior	Rational utility maximization for all agents	Standard game-theoretic assumption enabling equilibrium analysis
Information Structure	Complete system state visibility for all agents	Models connected vehicle environment with real-time data sharing
Demand Process	Independent EV arrivals with known distributions	Enables analytical tractability while capturing stochastic demand
Cost Structure	Linear additive costs across all components	Mathematical convenience while preserving key cost trade-offs
Capacity Modeling	Hard capacity constraints with binary occupancy states	Reflects physical charging port limitations accurately
Pricing Constraints	MCV prices bounded: $p_m \in [1.1, 2.0] \times p_{fcs}$	Ensures market competitiveness while covering operational costs
Spatial Uniformity	Homogeneous cost parameters within simulation area	Focuses analysis on strategic deployment rather than local variations

energy delivered: higher utilization spreads costs over more kWh, lowering cost per unit, while under-utilization raises the cost per unit.

In practice, the same charger or vehicle can look cheap or expensive depending on how much it is used. If an FCS port sits idle during long off-peak hours, its fixed costs are recovered over fewer charging sessions, making each kWh more costly. By contrast, an MCV can relocate to demand hotspots, keeping its utilization higher and its effective cost per kWh lower. This difference is the key reason why MCVs may be more attractive in volatile or peak-driven environments, while new FCS ports only make economic sense if they can be kept busy most of the day.

Two testable thresholds: The economic feasibility can be expressed through two complementary conditions, one for MCVs and one for FCS ports.

MCV sessions threshold: An MCV is economically viable only if it serves a sufficient number of charging sessions per day to cover its fixed costs, routing costs, and the margin over energy costs. The required annual energy to break even is given by

$$E_{\text{year,MCV}}^* = \frac{\text{CRF } C_{\text{MCV}} + F_{\text{MCV}} + c_{\text{km}} d_{\text{year}}}{p - (c_e + c_{\text{deg}})}, \quad (\text{S4})$$

where p is the sale price of energy. Expressed in sessions per day, the minimum number of daily sessions required for financial viability is

$$N_{\text{sess/day}}^* = \frac{E_{\text{year,MCV}}^*}{365 k_{\text{sess}}}, \quad (\text{S5})$$

with k_{sess} as the average session size. If observed sessions per MCV per day exceed $N_{\text{sess/day}}^*$, the MCV is feasible under the assumed price and cost parameters.

FCS utilization threshold: A new FCS port is justified only if its average utilization exceeds a break-even level. For a port with rated power P_{port} and observed utilization U_{port} , the annual energy is

$$E_{\text{year}}^{\text{FCS}} = 365 \times 24 \times P_{\text{port}} \times U_{\text{port}}. \quad (\text{S6})$$

Rearranging the leveled cost expression yields the break-even utilization:

$$U_{\text{min}} = \frac{\text{CRF } C_{\text{FCS}} + F_{\text{FCS}}}{(p_{\text{FCS}} - c_e) 365 \cdot 24 \cdot P_{\text{port}}}. \quad (\text{S7})$$

If the observed utilization U_{port} is below U_{min} , then the port does not break even and becomes loss-making because its fixed costs are spread over too few kilowatt-hours [4]. In our testbed, each new FCS represents a 10-port expansion, so the under-utilization risk is amplified.

From 8-hour to a representative day (TOU weighting): Urban demand concentrates in peak hours, with moderate daytime shoulders and long overnight off-peak. To scale 8 h results to a day without re-simulation, we convert hours into *peak-equivalent* hours using weights tied to observed arrival-rate ratios. This mirrors the concept of Equivalent Full-Load Hours (EFLH) used in energy systems to represent non-uniform operation as the number of hours at rated load [5]. Let w_{pk} , w_{nm} , and w_{op} denote the relative intensities in peak, normal, and off-peak. The 8 h window has effective hours

$$H_{eff}^{(8)} = h_{pk}^{(8)} w_{pk} + h_{nm}^{(8)} w_{nm} + h_{op}^{(8)} w_{op},$$

and a representative day has H_{eff}^{day} constructed analogously. The daily scaling factor is

$$\text{Scale factor} = \frac{H_{eff}^{day}}{H_{eff}^{(8)}}. \quad (S8)$$

This factor multiplies any 8 h energy to obtain a realistic daily value that reflects peak, shoulder, and off-peak intensity.

Decision rule: The two thresholds together provide a clear operational rule. Given (S2)-(S8):

Prefer MCV if the observed sessions per day per MCV are greater than or equal to the *break-even sessions* $N_{sess/day}^*$ from (S5). This means the MCV delivers enough charging sessions to cover its fixed and routing costs, maintaining feasibility through relocation to peaks and hotspots.

Expand FCS only if the observed utilization of each port U_{port} is greater than or equal to the *break-even utilization* U_{min} from (S7). If utilization is lower, the fixed capital and operating costs are spread over too few kilowatt-hours, which raises $LCOC_{FCS}$ in (S3) and makes the new capacity loss-making. In contrast, mobility allows MCVs to keep E_{year}^{MCV} high in (S2), since they can continuously reposition to demand-dense areas and maintain economic viability.

This theoretical foundation establishes the economic framework by defining how one-time investments are annualized, how costs are translated into a levelized cost per kWh, and how break-even thresholds are expressed for MCVs (in sessions per day) and FCS ports (in utilization levels). These definitions serve as reference points so that, in the results section, the empirical values can be directly compared against the theoretical break-even conditions.

S2.2. Economic feasibility and break-even: empirical evaluation

We adopt standard techno-economic parameters for EV charging that follow two benchmark LCOC studies, which compile charger hardware, installation, operations and management, financing, efficiency, and utilization assumptions across regions [6, 7]. Building on this, we now instantiate the economic thresholds from Section S2.1 on the observed operations and price-cost scenarios. The calculations compare (i) the MCV daily sessions threshold N^* from (S4)-(S5)

against the observed throughput, and (ii) the FCS utilization floor U_{min} from (S7) against the observed utilization. The observed *sessions per MCV per day* is 40.4, obtained by scaling the 8 h window to a representative day via (S8) to get $E_{day}^{MCV} = 2,021.87 \text{ kWh/MCV/day}$ and dividing by the average session size $k_{sess} = 50 \text{ kWh}$. The observed *port utilization* is $U_{port} = 0.472$, computed from (S6) as $E_{year}^{FCS} / (365 \cdot 24 \cdot P_{port})$ with $E_{year}^{FCS} = 0.414 \text{ GWh/port/year}$ and $P_{port} = 100 \text{ kW}$. **Units:** Prices are in \$/kWh, the average session size k_{sess} is in kWh, and rated charger power P_{port} is in kW.

Price-margin sensitivity: For MCVs, higher p_{MCV} – ($c_e + c_{deg}$) lowers N^* and $LCOC_{MCV}$; for FCS, higher p_{FCS} – c_e lowers U_{min} and raises 10-port headroom. The numerical effects under Scenarios A-C are in Table S3.

Scenario-driven insights: Scenario A (standard energy cost): FCS passes with $U_{min} = 0.312 < 0.472$ and positive 10-port headroom. MCV dispatch is attractive for peaks since 40.4 exceeds N_{oper}^* , but a new MCV does not clear the full-cost threshold. Scenario B (lower energy cost): FCS is most attractive and a new MCV is full-cost viable at $p_{MCV} = 0.39$ because 40.4 exceeds $N_{full}^* = 32.4$. Scenario C (higher energy cost): FCS does not pass at today's utilization and MCV dispatch is only attractive at higher p_{MCV} ; a new MCV is not full-cost viable.

Before the concluding rules, two practical notes. First, CAPEX varies: FCS costs are location dependent and MCV costs are brand dependent. Profitability is margin driven in both cases. For MCVs it is difficult to make profit on MCV operations alone unless energy is procured at low c_e and sold at a sufficiently high p_{MCV} . For FCS, the energy gross margin $p_{FCS} - c_e$ is decisive: as a rule of thumb, a sustained margin near 20% relative to the tariff can support viability; if the margin erodes, even high utilization can become loss making. Our experiments capture these dynamics through the margin sweeps in Table S3. Second, if annual routing increases to $d_{year} = 13,154 \text{ km}$, then N_{oper}^* increases by 50% (for example 12.0 to 18.0 and 8.7 to 13.1), N_{full}^* increases by 10.6% across all scenarios, and $LCOC_{MCV}$ increases by 1.19 c/kWh. LCOC represents the average all-in cost per kWh, combining annualized fixed cost, routing, energy, and degradation; values below the selling price indicate headroom for profit.

Clear operator rules under the present assumptions:

1. Add one 10-port FCS if the expected annual contribution margin across the ten ports exceeds $10 (CRFC_{FCS} + F_{FCS}) = \$163,900$. At $U_{port} = 0.472$, the expected contributions are \$248,400 (Scenario A), \$372,600 (Scenario B), and \$124,200 (Scenario C); add in A and B, do not add in C.
2. Prefer MCV dispatch whenever expected sessions per MCV per day exceed the operating threshold N_{oper}^* in Panel A. With 40.4 sessions today this holds in all cases except Scenario C at $p_{MCV} = 0.33$.

Table S2

Inputs, anchors, and scenarios used to compute MCV sessions thresholds N^* (S4)-(S5), the FCS utilization floor $U_{\min}$ (S7), and LCOC (S2)-(S3). Anchors include the daily scaling from (S8), sessions per MCV per day = $E_{\text{day}}^{\text{MCV}}/k_{\text{sess}}$, and U_{port} from (S6).

Item	Symbol	Value (units)	Used for
MCVs in fleet	m	6	8 h anchor
Daily scale factor	$\bar{H}_{\text{eff}}/H_{\text{eff}}^{(8)}$	2.403	map 8 h to day
Sessions per MCV per day	observed	40.4 sessions/day	compare vs N^*
Per MCV annual energy	$E_{\text{year}}^{\text{MCV}}$	0.738 GWh/MCV/y	LCOC _{MCV} denom
Per port annual energy	$E_{\text{year}}^{\text{FCS}}$	0.414 GWh/port/y	LCOC _{FCS} , fixed cost cap
Observed port utilization	U_{port}	0.472	compare vs $U_{\min}$
Real discount rate	r	0.07	CRF in (S1)
Lifetimes	$(n_{\text{MCV}}, n_{\text{FCS}})$	8 y, 10 y	with r , set CRF
CRF	(MCV, FCS)	0.167, 0.142	numerators in LCOC
MCV CAPEX	C_{MCV}	300,000 \$/MCV	fixed cost
MCV fixed OPEX	F_{MCV}	15,000 \$/MCV/y	fixed cost
FCS CAPEX	C_{FCS}	80,000 \$/port	fixed cost
FCS fixed OPEX	F_{FCS}	5,000 \$/port/y	fixed cost
Rated power	P_{port}	100 kW	in $U_{\min}$
Avg. session size	k_{sess}	50 kWh	in N^*
Routing cost	c_{km}	2.00 \$/km	MCV numerator
Annual routing	d_{year}	8,769 km (base); 13,154 km (+50%)	sensitivity on N^*
Annualized fixed cost	$\text{CRFC} + F$	65,240 \$/MCV/y; 16,390 \$/port/y	in LCOC, N^* , $U_{\min}$
Scenario A (standard c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.33, 0.24, 0.04) \$/kWh	margins: FCS 0.06, MCV 0.05
Scenario A (standard c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.36, 0.24, 0.04) \$/kWh	margins: FCS 0.06, MCV 0.08
Scenario A (standard c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.39, 0.24, 0.04) \$/kWh	margins: FCS 0.06, MCV 0.11
Scenario B (lower c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.33, 0.21, 0.04) \$/kWh	margins: FCS 0.09, MCV 0.08
Scenario B (lower c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.36, 0.21, 0.04) \$/kWh	margins: FCS 0.09, MCV 0.11
Scenario B (lower c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.39, 0.21, 0.04) \$/kWh	margins: FCS 0.09, MCV 0.14
Scenario C (higher c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.33, 0.27, 0.04) \$/kWh	margins: FCS 0.03, MCV 0.02
Scenario C (higher c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.36, 0.27, 0.04) \$/kWh	margins: FCS 0.03, MCV 0.05
Scenario C (higher c_e)	$(p_{\text{FCS}}, p_{\text{MCV}}, c_e, c_{\text{deg}})$	(0.30, 0.39, 0.27, 0.04) \$/kWh	margins: FCS 0.03, MCV 0.08

3. Add a new MCV only if expected sessions exceed the full-cost threshold N_{full}^* in Panel A. With 40.4 sessions today this holds only in Scenario B at $p_{\text{MCV}}=0.39$.

Thus, this subsection ties observed operations to clear economic thresholds. Updating Table S2 for a new city automatically regenerates Table S3 and the same decision logic.

S3. Simulation Environment and Setup

Consistent with the main manuscript, meeting points are the preselected candidate meeting locations where MCVs can serve EVs, derived from the Dynamic Rank-Based Meeting-Point Ranking procedure. At time zero, no MCVs are deployed to any meeting point, and the simulation is initialized with 30 EV requests to provide a realistic initial load, as shown in Figure S1(b). This initial configuration represents a snapshot of the overall service area, capturing the status of all FCSs, MCVs, and EVs at the same instant. Starting with all stations either completely empty or excessively crowded would bias the outcomes toward specific scenarios; therefore, a balanced and realistic starting point is adopted. The allocation of EVs to specific FCSs, indicated by blue dotted lines, is shown in Figure S1(c), demonstrating that the available FCSs can adequately serve the initial EV demand. While 3 FCSs form our baseline, we have also

evaluated scenarios with 4, 5, and 6 FCSs. The locations of additional FCSs are determined using the **Dynamic Rank-Based Final Meeting Point Ranking** algorithm, while the capacity and number of parallel charging ports remain unchanged in all scenarios. The simulation spans 8 hours, covering normal, peak, and off-peak demand periods, during which more than 500 EV requests are served, with arrival rates varying dynamically over time. This ensures the problem setting reflects realistic variability, where the number of EVs, their arrival patterns, and their required charging levels change continuously. The simulation logs minute-by-minute FCS and MCV statuses, including queue lengths, utilization rates, and EV assignments. Static snapshots and videos (tracking EV arrivals, cumulative assignment mapping, and full dynamics) were generated to visually capture the system evolution and can be accessed on the same GitHub repository.

S4. Complexity Details

This framework is designed for online operation, where deployment and pricing are updated at fixed control intervals. We separate costs into (i) an offline ranking stage that prepares a compact candidate set of meeting points and (ii) the online control cycle that repeats during simulation. Let $|I|$ be the number of MCVs, $|J|$ the number of FCSs, $|\mathcal{M}|$ the number of potential meeting points, and $|\mathcal{T}_{\text{rank}}|$

Table S3

Derived results from Table S2. Panel A gives MCV operating and full-cost sessions thresholds and LCOC at baseline routing $d_{\text{year}}=8,769$ km. Panel B gives the FCS utilization floor, LCOC, and 10-port headroom at $U_{\text{port}}=0.472$ and $P_{\text{port}}=100$ kW.

Panel A: MCV thresholds and cost

Scenario	p_{MCV} [\$/kWh]	MCV margin [\$/kWh]	N_{oper}^* [sess/day]	N_{full}^* [sess/day]	LCOC_{MCV} [\$/kWh]	Meets full-cost at 40.4?
A	0.33	0.05	19.2	90.7	0.392	No
A	0.36	0.08	12.0	56.7	0.392	No
A	0.39	0.11	8.7	41.2	0.392	No
B	0.33	0.08	12.0	56.7	0.362	No
B	0.36	0.11	8.7	41.2	0.362	No
B	0.39	0.14	6.9	32.4	0.362	Yes
C	0.33	0.02	48.1	226.8	0.422	No
C	0.36	0.05	19.2	90.7	0.422	No
C	0.39	0.08	12.0	56.7	0.422	No

$N_{\text{oper}}^* = \frac{c_{\text{km}} d_{\text{year}}}{365 k_{\text{sess}} [p_{\text{MCV}} - (c_e + c_{\text{deg}})]}$ considers only operating terms for dispatch, with capital and fixed operations and maintenance costs held constant.

N_{full}^* adds $\text{CRFC}_{\text{MCV}} + F_{\text{MCV}}$ to the numerator. $\text{LCOC}_{\text{MCV}} = \frac{\text{CRFC}_{\text{MCV}} + F_{\text{MCV}} + c_{\text{km}} d_{\text{year}}}{E_{\text{year}}^{\text{MCV}}} + c_e + c_{\text{deg}}$,

with $E_{\text{year}}^{\text{MCV}}=0.738$ GWh. With +50% routing, N_{oper}^* rises by 50%, N_{full}^* by 10.6%, and LCOC_{MCV} by 1.19 c/kWh.

Panel B: FCS utilization floor and 10-port economics

Scenario	FCS margin [\$/kWh]	U_{min} [-]	LCOC_{FCS} [\$/kWh]	10-port headroom vs \$163,900 [\$/y]	Viable at $U=0.472$?
A	0.06	0.312	0.280	84,500	Yes
B	0.09	0.208	0.250	208,700	Yes
C	0.03	0.624	0.310	-39,700	No

$U_{\text{min}} = \frac{\text{CRFC}_{\text{FCS}} + F_{\text{FCS}}}{(p_{\text{FCS}} - c_e) 365 \cdot 24 \cdot P_{\text{port}}}$ with $P_{\text{port}}=100$ kW.

Headroom compares $10(p_{\text{FCS}} - c_e) E_{\text{year, port}}^{\text{FCS}}$ with $10(\text{CRFC}_{\text{FCS}} + F_{\text{FCS}})=\$163,900$, using $E_{\text{year, port}}^{\text{FCS}}=0.414$ GWh.

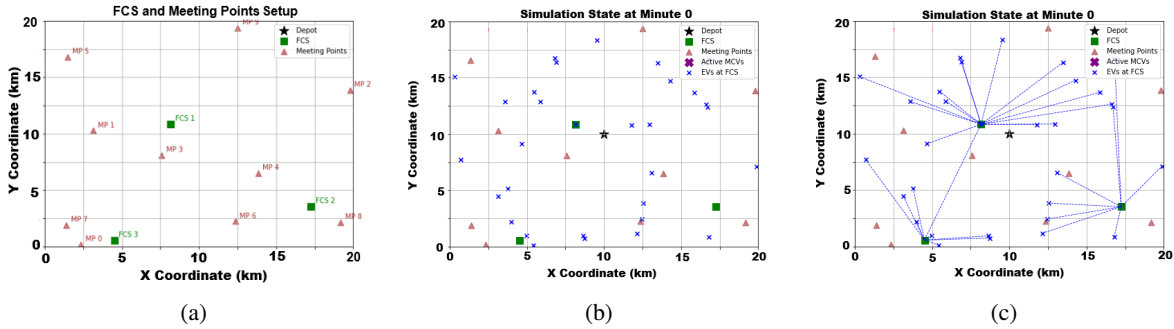


Figure S1: (a) Simulation area showing FCS and meeting point locations; (b) EV charging request locations; (c) EV allocation initialized to FCSs based on the Stackelberg game-based cost minimization

the number of ranking days. Let $\mathcal{K}_t$ be the EV arrivals in cycle t , $A_t := |\mathcal{K}_t|$ the number of active EV decisions in that cycle, $M_c \subseteq \mathcal{M}$ the pruned candidate set passed to the optimizer which is a set of meeting points that the optimizer actually considers in the leader decision step, and $I_d \leq \min(|M_c|, |I|)$ the number of distinct active meeting-point options visible to drivers in that cycle.

Offline rank preparation: For each day and each EV, the procedure compares the best FCS option to each meeting point, then aggregates and sorts the meeting points. The total cost is

$$\mathcal{O}\left(\sum_{d \in \mathcal{T}_{\text{rank}}} |\mathcal{K}_d| (|J| + |\mathcal{M}|)\right) + \mathcal{O}(|\mathcal{T}_{\text{rank}}| |\mathcal{M}| \log |\mathcal{M}|),$$

which is highly parallelizable across EVs and across days. This stage is run once to produce M_c (for example, the top 10-30 points).

Online follower choices: Each arriving EV evaluates effective costs over all FCSs and the currently deployed MCV options. With rolling estimates for waits and distances, the per-EV work is $\mathcal{O}(|J| + I_d)$, hence $\mathcal{O}(A_t(|J| + I_d))$ per cycle.

Online monitoring and triggers: Queue and utilization checks over FCSs and deployed MCVs are $\mathcal{O}(|J| + I_d)$ per evaluation, which is dominated by the follower work above.

Leader deployment decision: The exact leader step decides deployment over M_c with binary variables $\{x_{im}\}$ and

Table S4

Symbols and Variables Used in Runtime and Scalability Analysis

Symbol	Description
Basic Sets and Cardinalities	
$ I $	Number of Mobile Charging Vehicles (MCVs)
$ J $	Number of Fixed Charging Stations (FCSs)
$ \mathcal{M} $	Number of potential meeting points
$ \mathcal{T}_{\text{rank}} $	Number of ranking days used in offline preparation
$\mathcal{K}_t$	Set of EV arrivals in control cycle t
$ \mathcal{K}_d $	Number of EVs on ranking day d
Control Cycle Variables	
A_t	Number of active EV decisions in cycle t (i.e., $A_t = \mathcal{K}_t $)
M_c	Pruned candidate set of meeting points passed to optimizer ($M_c \subseteq \mathcal{M}$)
$ M_c $	Size of candidate meeting point set (typically 10-30)
I_d	Number of distinct active meeting-point options visible to drivers in a given cycle ($I_d \leq \min(M_c , I)$)
t	Control cycle index
d	Ranking day index
Decision Variables	
x_{im}	Binary variable: 1 if MCV i is deployed to meeting point m , else 0
B	Number of binary variables in leader MILP ($B = I \cdot M_c + \mathcal{O}(I)$)
Multi-Objective Optimization	
S	Set of candidate deployment configurations for Pareto analysis
$ S $	Number of candidate configurations evaluated
Scalability Parameters	
N_{cfg}	Number of unique system configurations tested
R	Number of replications for each configuration
U_t	Number of temporarily unserved EVs in cycle t
Computational Complexity Terms	
$\mathcal{O}(\cdot)$	Big-O notation for computational complexity
$\text{SolveTime}(\cdot)$	Wall-clock time to solve MILP with given number of binary variables
$\sum_t$	Summation over control cycles t
$\sum_d$	Summation over ranking days d

a small set of auxiliary binaries (for example, return decisions). The number of binaries scales as

$$B = |I| |M_c| + \mathcal{O}(|I|).$$

This generalizes capacitated facility location and is NP-hard, with worst-case exponential dependence on B . In practice, three choices keep cycle times stable: pruning to a small $|M_c|$, a single-cycle horizon, and warm starts from the previous solution (optionally with a small optimality gap or time limit).

The only online combinatorial component is the leader MILP; its wall-clock is kept small in practice by bounding $|M_c|$, warm-starting from the previous cycle, and using a modest optimality gap or time limit. In the simulation, this is implemented as follows: (i) **Pruning** meeting points to a

small candidate set ($|M_c|$ typically 10–30) during the offline ranking phase; (ii) adopting a **single-cycle horizon** to avoid solving for long-term schedules; (iii) **warm-starting** the MILP with the prior cycle's deployment to reuse and adjust existing assignments; and (iv) setting a **modest optimality gap or time limit** so that the solver halts when a near-optimal solution is reached. These measures ensure that although the MILP is NP-hard in theory, the problem size in practice remains tractable, allowing deployment decisions to be computed comfortably within the 5-minute control interval used in the experiments.

State updates: With cached priority queues for reassignment and return-to-depot checks, housekeeping is $\mathcal{O}(A_t)$ per cycle.

Multi-objective Pareto selection: Let S denote the set of candidate deployment configurations evaluated on a fixed number of objectives (e.g., revenue, utilization, unserved EVs, average wait). We extract the non-dominated set with a straightforward pairwise dominance check, which costs $\mathcal{O}(|S|^2)$, followed by a prioritized tie-break within the non-dominated set in $\mathcal{O}(|S| \log |S|)$. This step is offline and parallelizable across configurations.

Overall bound and scaling behavior: Over one horizon (e.g., a day), the combined work is summarized as follows. $\mathcal{T}_{\text{rank}}$ denotes the set of ranking days; I_d is the number of *distinct active meeting points* visible to drivers in a cycle (so $I_d \leq \min(|M_c|, |I|)$); $B = |I| |M_c| + \mathcal{O}(|I|)$ is the number of leader binaries; and S is the set of candidate configurations considered in the Pareto analysis. Two clarifications that guided our design choices. (i) The online linear terms also subsume several per-cycle routines that do not change the asymptotic order: (a) MCV price updates $\sum_t \mathcal{O}(|M_c|)$, (b) return/SoC policy checks $\sum_t \mathcal{O}(|I|)$, and (c) periodic reassignment of temporarily unserved EVs $\sum_t \mathcal{O}(U_t(|J| + |M_c|))$. Each is linear in the number of vehicles/options and is dominated by the EV-choice scans $\sum_t \mathcal{O}(A_t(|J| + I_d))$. (ii) The only online combinatorial component is the leader MILP; in practice we bound $|M_c|$, warm-start from the previous cycle, and impose a modest optimality gap or time limit, which keeps wall-clock well within the control interval. The offline layers scale as $\mathcal{O}(\sum_{d \in \mathcal{T}_{\text{rank}}} |\mathcal{K}_d|(|J| + |\mathcal{M}|))$ for ranking plus $\mathcal{O}(|S|^2)$ for Pareto front computation (with $\mathcal{O}(|S| \log |S|)$ tie-breaks), and are parallelizable. Finally, when stress-testing alternative FCS/MCV configurations and robustness replications, the offline workload scales multiplicatively. The total computational effort is proportional to $N_{\text{cfg}} \cdot R$, where N_{cfg} represents the number of unique system configurations tested and R is the number of replications for each. This highly parallelizable outer loop does not affect online responsiveness, and our implementation meets a 5-minute control cycle.

$$\begin{aligned}
 & \underbrace{\sum_t \mathcal{O}(A_t(|\mathcal{J}| + I_d))}_{\text{EV choices \& triggers}} + \underbrace{\mathcal{O}\left(\sum_{d \in \mathcal{T}_{\text{rank}}} |\mathcal{K}_d|(|\mathcal{J}| + |\mathcal{M}|)\right)}_{\text{Ranking: comps}} \\
 & + \underbrace{\sum_t \text{SolveTime}(B = |\mathcal{I}| |\mathcal{M}_c| + \mathcal{O}(|\mathcal{I}|))}_{\text{Leader MILP (NP-hard)}} + \underbrace{\sum_t \mathcal{O}(A_t)}_{\text{State updates}} \\
 & + \underbrace{\mathcal{O}(|\mathcal{T}_{\text{rank}}| |\mathcal{M}| \log |\mathcal{M}|)}_{\text{Ranking: sort}} + \underbrace{\mathcal{O}(|\mathcal{M}_c|)}_{\text{Placement}} + \underbrace{\mathcal{O}(|\mathcal{S}|^2)}_{\text{Pareto: front}} \\
 & + \underbrace{\mathcal{O}(|\mathcal{S}| \log |\mathcal{S}|)}_{\text{Pareto: tie-break}}. \quad (\text{S9})
 \end{aligned}$$

All online terms are linear in arrivals and near-linear in options per EV (i.e., $|\mathcal{J}| + I_d$); the only online combinatorial component is the leader MILP, whose wall-clock is kept small in practice by bounding $|\mathcal{M}_c|$, warm-starting, and using a modest optimality gap or time limit. The offline ranking and Pareto layers scale with $|\mathcal{T}_{\text{rank}}|$ and $|\mathcal{S}|$, respectively, and are parallelizable. Consequently, because the computationally intensive components are handled offline or strategically constrained, the framework's effective online complexity tends toward linear, making it feasible for real-world deployment. The actionable insights derived from the proposed methodology are therefore not just theoretically optimal but also practically deployable, offering flexible, data-driven guidelines for policymakers and industry stakeholders.

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